

LAB : 10

Title : More About Functions in C++

► Objectives

- Understand function call by value.
- Understand function call by reference.
- Learn about default parameter values in functions.
- Apply these concepts in practical C++ programs.

► Theory

Call by Value: A copy of the variable is passed to the function. Changes inside the function do not affect the original variable.

Call by Reference: The actual memory address is passed. Changes inside the function directly affect the original variable.

Default Parameters: Functions can have default values for parameters, which are used if no argument is provided during the call.

► Lab Tasks

Task 1: Function with Parameters

```
#include <iostream>
```

```
// include input/output library
```

```
using namespace std;
```

```
// use standard namespace
```

```
void multiply(int num1, int num2) {           // function with two integer parameters
    int product = num1 * num2;               // calculate product of num1 and num2
    cout << "Product = " << product;        // display result inside function
}

int main() {                                 // main function starts
    multiply(6, 7);                           // call multiply function with two numbers
    return 0;                                // end program
}
```

Task 2: Function with Return Value

```
#include <iostream>                          // include input/output library
using namespace std;                         // use standard namespace

int getSquare(int number) {                  // function with one integer parameter
    int square = number * number;           // calculate square of the number
    return square;                          // return the result
}

int main() {                                // main function starts
    int result = getSquare(9);               // call function and store returned value
    cout << "Square = " << result;          // display the square
    return 0;                               // end program
}
```

► Results

- Task 1 displayed: Product = 42 (for inputs 6 and 7).
- Task 2 displayed: Square = 81 (for input 9).

► Conclusion

In this lab, I learned:

- How to pass parameters to functions.
- How to return values from functions.
- The difference between call by value and call by reference.
- How default parameters simplify function calls.

Functions make programs modular, reusable, and efficient.